

3.6 Carboxylic Acids & Derivatives

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.6 Carboxylic Acids & Derivatives
Difficulty	Easy

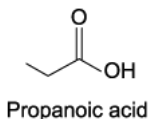
Time allowed: 20

Score: /10

Percentage: /100

Question 1

The natural bacterial fermentation of milk produces a compound called propanoic acid. Propanoic acid is partly responsible for the flavour of Swiss cheese.



Which starting materials could be used to produce propanoic acid?

- 1 $\text{CH}_3\text{CH}_2\text{CN}$
- 2 $\text{CH}_3\text{CH}_2\text{CHO}$
- 3 $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]

Question 2

Synthesis of the pharmaceutical drug aspirin requires ethanoic acid, $\text{CH}_3\text{CO}_2\text{H}$.

Which processes can be used to make ethanoic acid?

- 1 Oxidation of ethanol
- 2 Oxidation of ethanal
- 3 Hydrolysis of ethanenitrile

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]

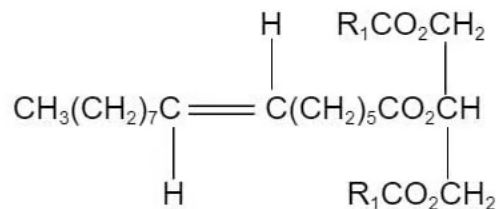
Question 3

'Trans fats' can be found in some vegetable oils and are known to lead to increased blood cholesterol levels.

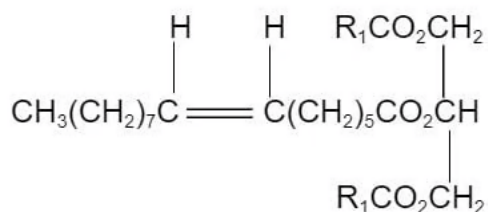
In the diagrams below, R_1 and R_2 are different hydrocarbon chains.

Which diagram shows an optically active 'trans fat'?

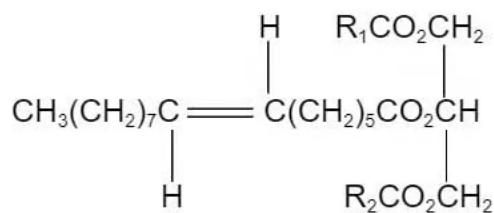
A.



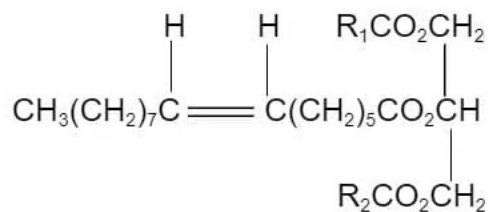
B.



C.



D.



[1 mark]

Question 4

Methanoic acid is one of the two reactants used to produce an ester with the molecular formula $C_5H_{10}O_2$.

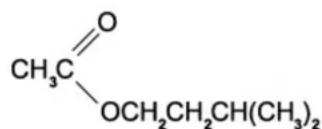
How many isomeric esters, including structural isomers and stereoisomers, could be made with this molecular formula?

- A. 3
- B. 4
- C. 5
- D. 6

[1 mark]

Question 5

The structure of ester **E** is shown in the diagram below.



E is hydrolysed in the presence of hydrochloric acid.

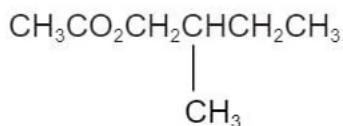
Which are the products of this hydrolysis?

- A. CH_3CO_2H and $(CH_3)_2CHCH_2CH_2CHO$
- B. CH_3CO_2H and $(CH_3)_2CHCH_2CH_2OH$
- C. $CH_3COC/$ and $(CH_3)_2CHCH_2CH_2OH$
- D. CH_3CHO and $(CH_3)_2CHCH_2CH_2OH$

[1 mark]

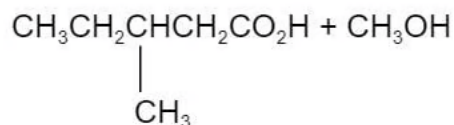
Question 6

The ester in the diagram below has an odour of banana.

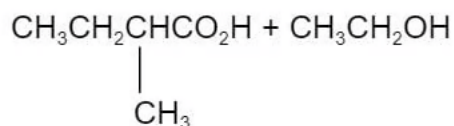


Which pair of reactants can be used to produce this ester under suitable conditions?

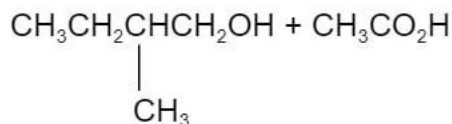
A.



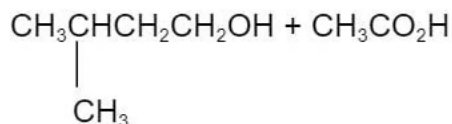
B.



C.



D.



[1 mark]

Question 7

Which reaction does **not** produce propanoic acid?

- A. Heating propanenitrile under reflux with dilute sodium hydroxide.
- B. Heating propanenitrile under reflux with dilute sulfuric acid.
- C. Heating propanal under reflux with acidified sodium dichromate (VI).
- D. Heating propanol under reflux with acidified sodium dichromate (VI).

[1 mark]

Question 8

Propionitrile, $\text{CH}_3\text{CH}_2\text{C} \equiv \text{N}$, is used to produce propanoic acid.

What are the conditions for this reaction?

- A. NaOH
- B. $\text{H}^+ / \text{Na}_2\text{Cr}_2\text{O}_7$
- C. NaBH_4
- D. $\text{H}^+ / \text{H}_2\text{O}$

[1 mark]

Question 9

Sodium lactate, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{Na}$, can be produced from lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CO}_2\text{H}$.

Which reagents, when used in excess, can be used in this reaction?

- 1 Na
- 2 NaOH
- 3 NaHCO_3

- A. 1, 2 and 3
- B. 1 and 2
- C. 2 and 3
- D. 1 only

[1 mark]

Question 10

The molecule $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ is used as a pineapple flavouring in sweets.

Which statements about this molecule are correct?

- 1 When this compound is heated with aqueous sodium hydroxide, the products are butan-1-ol and sodium ethanoate.
- 2 The name of this compound is ethyl butanoate.
- 3 This compound is a structural isomer of hexanoic acid.

A. 1, 2 and 3

B. 1 and 2

C. 2 and 3

D. 1 only

[1 mark]

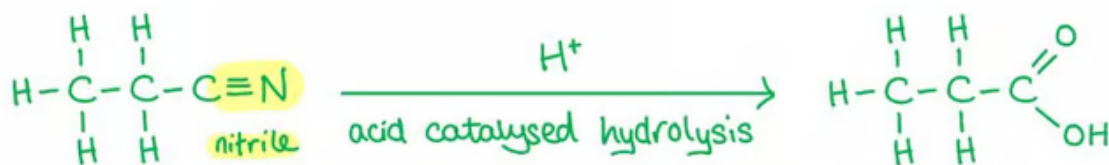
Model Answers: Easy

1

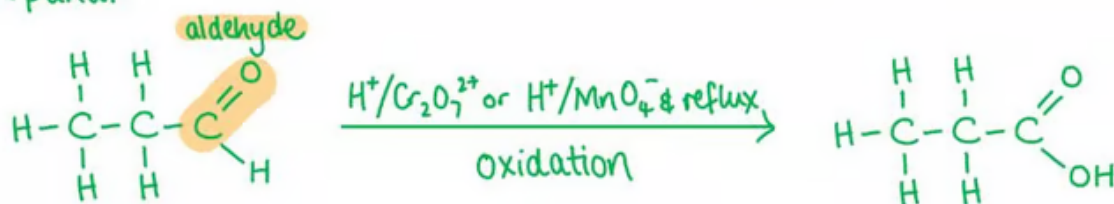
The correct answer is **A** because:

- Compound 1 is propanenitrile; nitriles are hydrolysed to carboxylic acids by heating the nitrile with dilute hydrochloric acid.
- Compound 2 is propanal; the propanal can be oxidised by heating and refluxing it with acidified potassium dichromate ($K_2Cr_2O_7$) or potassium permanganate ($KMnO_4$) solution.
- Compound 3 is propanol; the alcohol can be oxidised by heating and refluxing it with acidified potassium dichromate ($K_2Cr_2O_7$) or potassium permanganate ($KMnO_4$) solution.

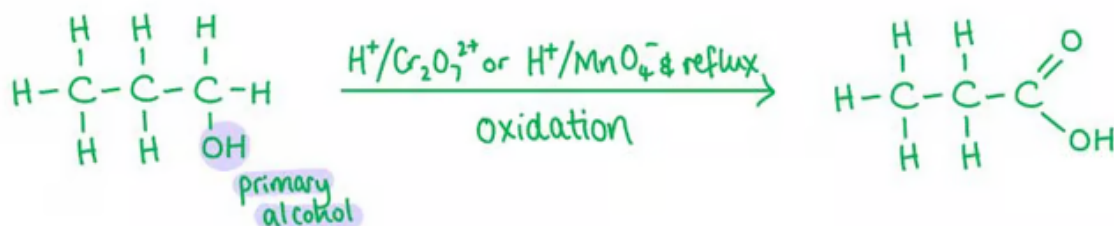
Propanenitrile:



Propanal:



Propan-1-ol:

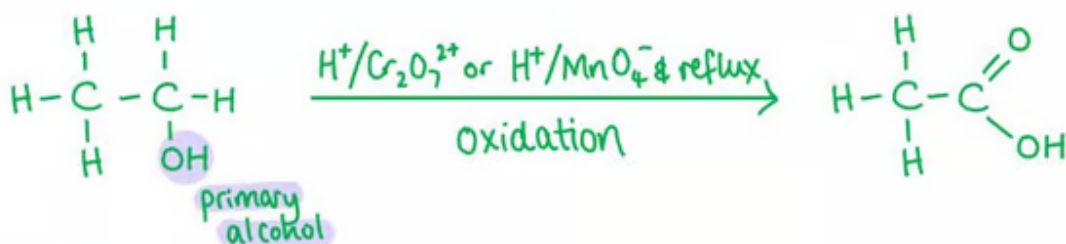


2

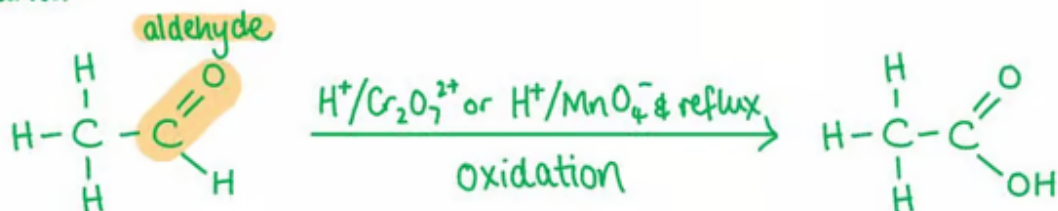
The correct answer is **A** because:

- Compound 1; the ethanol is heated with alkaline potassium manganate to produce ethanoic acid.
- Compound 2; ethanal; the ethanal can be oxidised by heating and refluxing it with acidified potassium dichromate ($K_2Cr_2O_7$) or potassium permanganate ($KMnO_4$) solution.
- Compound 3; ethanenitrile; nitriles can be oxidised by heating and refluxing it with acidified potassium dichromate ($K_2Cr_2O_7$) or potassium permanganate ($KMnO_4$) solution.

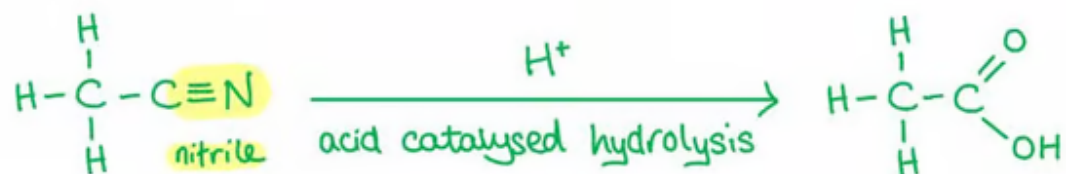
Ethanol:



Ethanal:



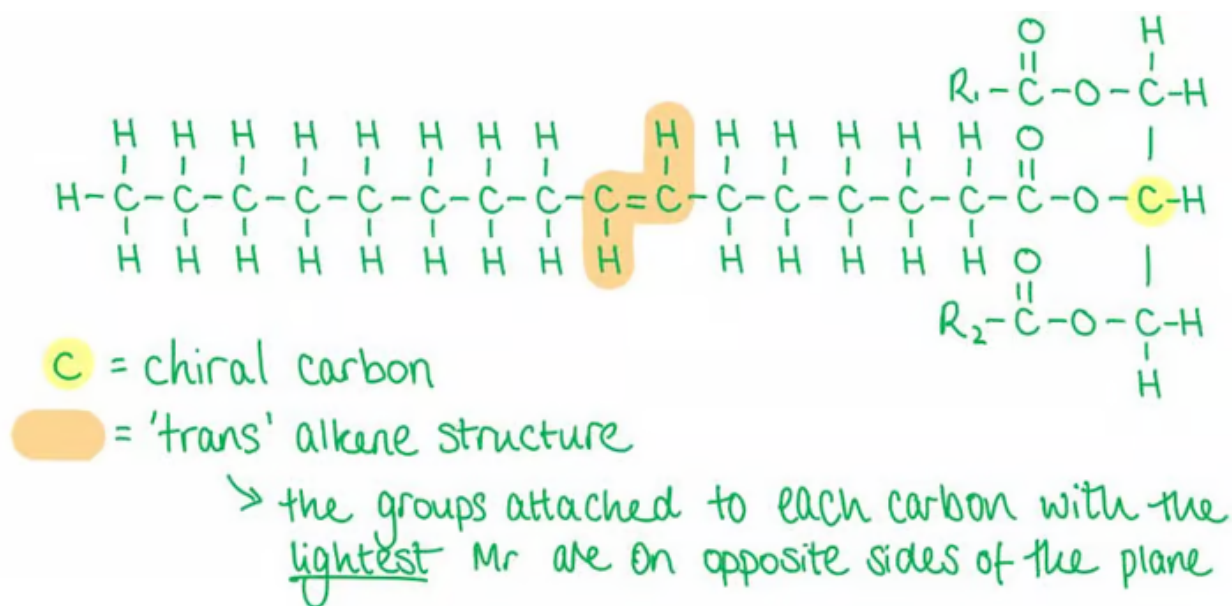
Ethanenitrile:



3

The correct answer is **C** because:

- 'Trans-fat' is a carboxylic acid with a long aliphatic chain that is either saturated or unsaturated.



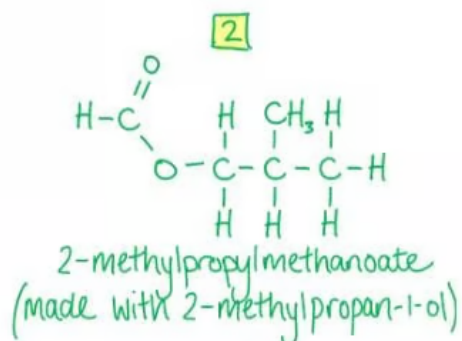
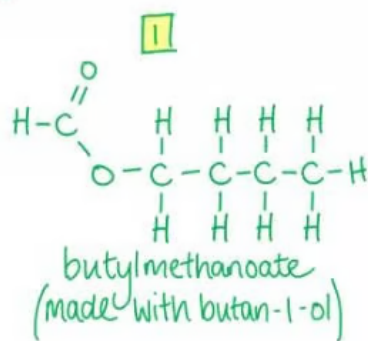
A is incorrect as this compound is not optically active as it does not have a chiral carbon.

B & D are incorrect as they are not 'trans' in structure as the hydrogens are on the same side of the double bond.

4

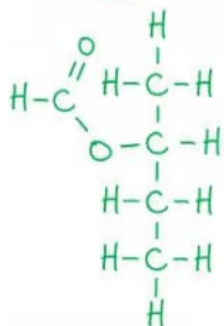
The correct answer is **B** because:

- Methanoic acid has the formula of HCOOH .
- When an ester is formed, it is HCOO- , the rest of the ester structure is $\text{-C}_4\text{H}_9$, this is a saturated 4 carbon chain.
- There are **4** structural isomers.



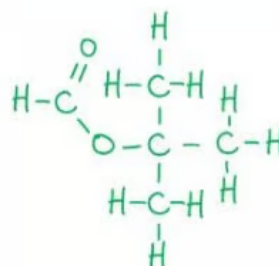
butylmethanoate
(made with butan-1-ol)

3



2-methylpropylmethanoate
(made with 2-methylpropan-1-ol)

4

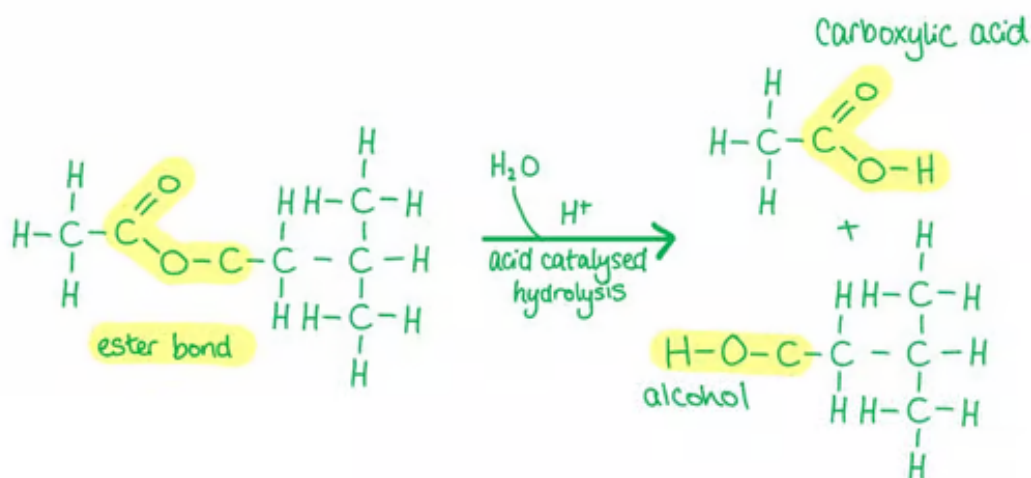


1-methylpropylmethanoate
(made with butan-2-ol)

1,1-dimethylpropylmethanoate
(made with 2-methylpropan-2-ol)

5

The correct answer is **B** because:



A is incorrect as the alcohol product is written as an aldehyde and has an extra extra carbon.

C is incorrect as the chlorine is a spectator ion in this reaction; this is a hydrolysis reaction.

D is incorrect as hydrolysing an ester gives a carboxylic acid and alcohol, CH_3CHO is an aldehyde and **not** a carboxylic acid.

6

The correct answer is **C** because:

- Esters are derived from carboxylic acids.
- The reactants shown in **C** are a simple two carbon carboxylic acid and branched alcohol and they would undergo an esterification reaction to produce the ester in the question.

A & B are incorrect as this is not the right combination of alcohol to the carboxylic acid. The ester needs to be a two carbon carboxylic acid and a branched alcohol. These options are the opposite.

D is incorrect as the methyl group is attached to the incorrect carbon.

7

The correct answer is **A** because:

- When propanenitrile is heated with sodium hydroxide under reflux, the products are sodium propanoate and ammonia.
- To produce the free carboxylic acid the products would need to be acidified.

B, C & D are incorrect as all of these reactions would produce propanoic acid.

8

The correct answer is **D** because:

- When propionitrile is heated under reflux with an acid it produces an ammonium salt and carboxylic acid.
- The propanoate ions react with the hydrogen ions from the acid to produce propanoic acid.

A is incorrect as while the nitriles can be hydrolysed with alkali under reflux, the resulting solution would need to be acidified to produce the carboxylic acid.

B is incorrect as acidified sodium dichromate is used to produce carboxylic acids from

propanol and propanal.

C is incorrect as sodium borohydride is a mild reducing agent and will not reduce nitro compounds under ordinary conditions.

9

The correct answer is **C** because:

- Sodium lactate is the sodium salt of lactic acid.
- This is essentially a neutralisation reaction as you are getting salt from the acid.
- This reaction will need an alkali or base with Na^+ in it.
- Therefore, only NaOH or NaHCO_3 can be used in excess to make sodium lactate from lactic acid.

A, B & D are incorrect as all of these options contain reagent 1 (Na). To form lactate, only deprotonation of the carboxylic acid should take place. However, if lactic acid is reacted with sodium (or any reactive metal), both the alcohol and carboxylic acid groups will react:

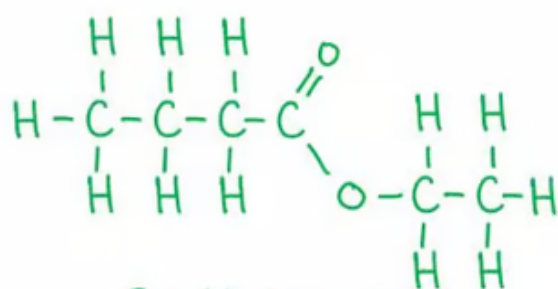
sodium + carboxylic acid $\rightarrow \text{RCOO}^-\text{Na}^+$ salt + H_2 (g) &

sodium + alcohol $\rightarrow \text{RCO}^-\text{Na}^+$ salt + H_2 (g)

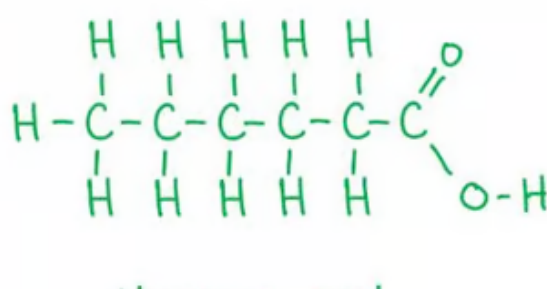
10

The correct answer is **C** because:

- The name of the compound would be ethyl butanoate, the structure shown in the diagram.
- This compound is a structural isomer of hexanoic acid, as shown in the diagram.



Ethyl butanoate



Hexanoic acid